

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (IT) Examination****March-April 2018****IT408 Service Oriented Architecture****Date: 06.04.2018, Friday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.****[07]**

- a. What is Service Oriented Architecture?
- b. Why SOAP is used for messaging in SOA?
- c. WSDL Stands for _____.
- d. “XML is strictly typed language”, true/false, Justify.
- e. What is REST based web service?
- f. Explain <message> element of SOAP?
- g. What is web3.0?

Q – 2.a Explain significance of ACID properties in SOA.**[04]****Q – 2.b Answer any two:****[10]**

- (i) Explain Syntax rules for XML.
- (ii) Discuss service dimensions with diagram.
- (iii) Write benefits of WSDL in SOAP based web services.

Q – 3.a Explain conceptual contents of service in SOA with Diagram.**[07]****OR****Q – 3.a Explain layered SOA Architecture with Diagram.****[07]****Q – 3.b Explain Enterprise perspective of SOA.****[07]****OR****Q – 3.b What is Software Architecture? Explain following Architectural principles.****[07]**

1. Reusability, 2. Loose Coupling, 3. Abstraction

SECTION – II**Q – 4 Answer the question below. [07]**

- a. What is Extensible Markup Language?
- b. Differentiate between business service and web service in SOA?
- c. “REST Web services are stateless”, true/false, justify.
- d. Why interoperation is more preferred than integration in SOA?
- e. How “Time-to-market” can be reduced by SOA?
- f. UDDI stands for _____.
- g. What is OWL in context of the Semantic Web?

Q – 5.a Answer any two. [10]

- (i) Differentiate between SOAP and REST web services.
- (ii) What is envelope in SOAP? Explain four fault types in SOAP.
- (iii) Explain motivation for OWL and following terms of OWL:
 1. Collection
 2. Properties

Q – 5.b Write short note on Choreography. [04]**OR****Q – 5.b Write short note on Orchestration. [04]****Q – 6.a Explain information architecture with figure. [04]****OR****Q – 6.a What are the benefits of SOA? Explain with example. [04]****Q – 6.b What is use of XSLT? Explain following elements of XSLT with example. [05]**

1. <xsl:if>
2. <xsl:sort>

Q – 6.c Write XML and DTD for Exam timetable as shown in figure 1. [05]**OR****Q – 6.c Write XML and XML Schema for Exam timetable as shown in figure 1. [05]****CHARUSAT EXAM TIMETABLE**

Sub code	Subject Name
IT709	Algorithms & Computational Complexity
IT710	Soft Computing

Figure 1
